



A novel integrated logistic regression model enhanced with recursive feature elimination and explainable artificial intelligence for dementia prediction

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ABSTRACT

Dementia is a major global health issue that significantly impacts millions of individuals, families, and societies worldwide, creating a substantial burden on healthcare systems. This study introduces a novel approach for predicting dementia by employing the Logistic Regression (LR) model, enhanced with Recursive Feature Elimination (RFE), applied to a unique dataset comprising 1000 patients, with 49.60% male and 50.40% female. The LR model, recognized for its simplicity and effectiveness in binary classification tasks, is optimized through RFE, a technique that iteratively eliminates less significant features to improve model performance. The model's effectiveness was assessed using comprehensive metrics, including accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and Kappa score. Furthermore, SHapley Additive exPlanations (SHAP) values were employed to increase the interpretability of the model, providing insights into the most influential features for dementia prediction. To address the issue of overfitting, a standardization technique was implemented, which enhanced the model's predictive performance. The findings of this study hold potential implications for early dementia detection, informing intervention strategies, and optimizing healthcare resource allocation.

1. Introduction

Dementia, a chronic and progressive disorder characterized by cognitive decline that exceeds normal age-related deterioration, poses a significant global health challenge, affecting millions of individuals worldwide [1]. The importance of addressing dementia lies in its profound impact on patients, their families, and healthcare systems. Early and precise prediction of dementia is crucial for patient care and disease management, as it can significantly alleviate the burden on individuals, families, and societies [2]. According to a recent report, there are currently 55 million people living with dementia worldwide,

and this number is projected to increase to 139 million by the year 2050 [3].

Clinical data plays an essential role in the early prediction and management of dementia. Accurate and comprehensive clinical data allows for better understanding, diagnosis, and treatment of the disease. By leveraging clinical data, healthcare providers can identify patterns and trends that may not be immediately apparent, leading to more effective interventions and improved patient outcomes [4,5].

Machine learning (ML), a subfield of artificial intelligence that involves the use of algorithms to analyze data, learn from it, and then

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make predictions, presents promising solutions for the early prediction of dementia [6]. Despite its potential, several challenges remain in applying ML to dementia prediction, such as limited data, overfitting, and the need for model interpretability [7].

This study aims to address these challenges by improving the LR model through the incorporation of RFE, a feature selection method that recursively removes features builds a model using the remaining attributes, and evaluates the model performance [8]. RFE helps in selecting the most informative features, thereby improving the efficiency and reducing the overfitting of the model.

Our study utilized a unique dataset named “Dementia Patient Health, Prescriptions ML Dataset”, available on Kaggle [9]. The dataset is a combination of data collected from various sources, including PUBMED, online research sources, NHS, Google Scholar, and consultations with healthcare professionals, specifically related to dementia. We address challenges such as overfitting and provide comprehensive evaluation metrics, including MCC and Kappa score, along with 10-fold cross-validation [10]. Furthermore, This current study utilize Explainable AI (xAI) techniques, specifically SHAP values, for model interpretability, providing insights into the features that are most influential in predicting dementia.

The main problem addressed in this study is the improvement of dementia prediction models to enhance early diagnosis and intervention. By leveraging advanced ML techniques and a robust dataset, this research aims to contribute to the field of dementia prediction in several significant ways:

- Predicting the presence or absence of dementia using a LR model.
- A feature extraction RFE technique was employed to overcome the overfitting challenges in the dementia dataset.
- Highlighting the explainable-ai to interpret the output as prior research focused on black box Machine Learning models

The following sections comprise the remainder of this paper: Section 2 presents a comprehensive review of previous research relevant to the current topic of investigation. Section 3 elucidates the research methodology undertaken to achieve the study objectives. Section 4 offers an in-depth analysis and discussion of the findings, culminating in the conclusion, which is followed by potential areas for further exploration in Section 5.

2. Literature review

Twait et al. (2023) explored the performance of machine learning algorithms in predicting all-cause dementia, using clinically accessible variables instead of relying on MRI predictors. Their data set includes 4793 participants from the AGES-Reykjavik Study, without dementia or mild cognitive impairment at baseline (average age: 76 years, 59% female). They feature a comprehensive set of variables, including demographics, clinical variables, medication use, lifestyle variables, cognitive assessment, and neuroimaging variables. They used machine learning algorithms including elastic net regression, random forest, support vector machine, and elastic net Cox regression. The study employed a ten-fold cross-validation, repeated ten times, and used upsampling to address imbalanced data. Machine learning algorithms performed comparably to LR across all three models [11].

Ortiz- Perez et al. (2023) developed an architecture using different deep-learning modules to process text transcriptions and audio recordings for early dementia symptom prediction. They used the DementiaBank Pitt Corpus dataset, containing audio recordings and their transcriptions for both healthy individuals and people with dementia. They used Convolutional Neural Networks (CNN) for audio classification and Transformers for text classification. The best results were achieved using text modality, reaching an accuracy of 90.36% in dementia detection. The audio model achieved 73.49% accuracy, and

the multimodal approaches did not significantly outperform the text-only model. Therefore, their limitation is that the multimodal models did not significantly improve results over the text-only model [12].

Franciotti et al. (2023) compared the prognostic performances of three supervised machine learning algorithms (Random Forest, Gradient Boosting, and eXtreme Gradient Boosting) in predicting MCI conversion to AD using multimodal biomarkers. Data were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database. The study involved categorizing MCI baseline data into neuropsychological features, structural Magnetic Resonance Imaging (sMRI)-related data, and AD-related biomarkers. Machine Learning (ML) algorithms implemented include Random Forest, Gradient Boosting, and eXtreme Gradient Boosting. The study used a ten-fold cross-validation, repeated ten times, and employed upsampling to address imbalanced data. The performance measures were calculated for individual data classes (neuropsychological, sMRI, and AD-related biomarkers) and their combinations. The highest accuracy (0.90) was reached using a combination of neuropsychological and AD-related biomarkers. The study highlighted that different ML algorithms have their strengths, with RF performing better on neuropsychological data and GB and XGB showing better performance with class combinations. Their limitation is the varying performance improvement after feature selection, indicating the complex nature of the datasets and the algorithms [13].

Kabir et al. (2023) detected dementia in Bangladesh, highlighting the challenge of recognizing dementia at an early stage due to limited awareness of the illness and its symptoms. They used a dataset consisting of 199 people with dementia and 175 healthy controls. The dataset included information such as age, gender, symptoms, and diagnosis. They used Decision Tree, Random Forest, K-Nearest Neighbor (KNN), and Support Vector Machine (SVM). The algorithms were trained and tested on the collected data to identify early signs of dementia. The KNN algorithm showed the most promising results with an accuracy of 96% in identifying dementia in its early stages. The research faced challenges due to the limited data available specifically for the early stages of dementia in Bangladesh [14].

Kim et al. (2023) developed an EEG-machine learning (ML) algorithm to detect brain amyloid pathology among SCD or MCI populations, validated with $A\beta$ PET scans. They collected data from 311 subjects, comprising 196 SCD (36 $A\beta$ -positive and 160 $A\beta$ -negative) and 115 MCI individuals (54 $A\beta$ -positive and 61 $A\beta$ -negative). EEG features were generated and selected, and six statistical analysis methods were employed to select important feature sets. The best-performing model showed a sensitivity of 90.9%, specificity of 76.7%, and accuracy of 82.9% in the combined MCI and SCD group. The model’s generalizability may be limited due to the mono-modality approach, and incorporating additional clinical data could improve robustness [15].

Li et al. (2023) investigated whether ML methods could improve the prediction of dementia in SVD patients compared to traditional statistical methods. They Utilized baseline demographics, vascular risk factors, cognitive scores, and MRI features related to SVD. 789 individuals from three cohorts (RUN DMC, SCANS, HARMONISATION), encompassing a range of SVD severity and ethnic diversity. Fifteen SVD-related features including demographic data, imaging features from MRI (like white matter lesion load, lacune count), and cognitive features. They used ml models Including Regularized Cox Proportional Hazards model (Reg_Cox), Random Survival Forests (RSF), Gradient Boosted Survival Trees (GBT), Support Vector Machine (SVM), and others. Internal validation through nested 5-fold cross-validation with hyperparameter tuning. ML models showed limited improvement over traditional statistical methods in survival analysis, regularized Cox/LR performed slightly better but not significantly so. The moderate number of participants might have constrained the training of more complex ML models [16].

Nyholm et al. (2024) examined if machine learning can predict dementia and identify which sleep disturbance factors significantly

Table 1
Few related studies.

Research	Method used	Limitation
[11] Twait et al. (2022)	Elastic Net Regression, RF, SVM,	Their study have also overfitting issue
[12] Ortiz-Perez et al.(2023)	Elastic Net Cox Regression Convolutional Neural Networks, Transformers for Text Classification, Multimodal Ensemble	Their used model achieves high accuracy, but the results are highly dependent on the characteristics of the DementiaBank Pitt Corpus dataset, which might not generalize well to other datasets or real-world scenarios where audio and textual data might vary significantly
[14] Kabir et al. (2023)	KNN, DT, SVM, RF	Their study has a overfitting issue
[15] Kim et al. (2023)	SVM, LR, KNN, NB, Random Forest , AdaBoost, Gradient Boosting Machine (GBM), XGBoost	There is a risk of overfitting, particularly because the study used multiple machine learning algorithms and extensive feature engineering, which could tailor the model too closely to the training data.
[13] Franciotti et al. (2023)	RF, Gradient Boosting, Extreme Gradient Boosting, SHAP	Their used boosting methods like GB and XGB can reduce bias and variance, these models are also prone to overfitting, especially with complex models and large datasets

impact dementia risk. They used data from the Swedish National Study on Aging and Care — Blekinge (SNAC-B) comprising older adults (60+ years) in Sweden. The study included 4175 participants, focusing on personal and sleep disturbance factors. They employed five machine learning algorithms: Gradient Boosting, LR, Gaussian Naive Bayes, Random Forest, and Support Vector Machine. The algorithms used a 10-fold stratified cross-validation approach for evaluation. The Brier score was used to check accuracy, and feature importance analysis was conducted to determine the factors impacting dementia. The study acknowledges limitations due to the focus on older adults in a specific geographic location (Blekinge, Sweden), which may affect the generalizability of the findings [17].

Cho et al. (2023) used machine learning models to predict the occurrence of BPSD among community-dwelling older adults with dementia. The study included 187 older adults with dementia for model training and 35 for external validation. Data types included demographic and health data, premorbid personality traits, actigraphy data (for monitoring sleep and physical activity), and symptom diary data (for tracking caregiver-perceived symptom triggers and daily occurrence of BPSD). Machine learning approaches employed included LR, random forest, gradient boosting machine, and support vector machine. Actigraphy was used to monitor sleep and physical activity levels, and a symptom diary was utilized to track caregiver-perceived symptom triggers and daily BPSD occurrence. The random forest models showed the highest area under the receiver operating characteristic curve (AUC) values for hyperactivity, euphoria/elation, and appetite/eating disorders. The generalizability of the model may be limited due to the mono-modality approach and the specific population studied [18].

Mostafaei et al. (2023) identified important variables associated with mortality risk in dementia patients using machine learning algorithms. They used data from the Swedish Registry for Cognitive/Dementia Disorders (SveDem). A longitudinal cohort of 28,023 dementia-diagnosed patients. Employed three machine learning algorithms with sparsity-inducing penalties — Logistic Regression (LR), Support Vector Machine (SVM), and Backpropagation Neural Networks (NN). Model performance was evaluated using metrics like Area-Under-ROC Curve (AUC). Identified twenty important variables for mortality risk prediction and fifteen for predicting time to death. Some potential variables might not have been included in the study, possibly affecting the comprehensiveness of the results [19].

Dhakal et al. (2023) created a predictive model for dementia using various supervised machine learning algorithms. It focuses on analyzing the dataset and features within it for dementia prediction. The data was obtained from the Open Access Series of Imaging Studies (OASIS) project. It included longitudinal MRI data of 150 subjects aged 60 to 96, both men and women, right-handed, categorized as nondemented, demented, and converted. Nine supervised machine learning

techniques were implemented: AdaBoost, Decision Tree, Extra Tree, Gradient Boost, K-NN, Logistic Regression, Naïve Bayes, Random Forest, and Support Vector Machine (SVM). The study involved data preparation, pre-processing (including feature selection using LASSO and Chi-square methods), training and testing split, and model evaluation. The SVM algorithm achieved the highest accuracy of 96.77% with full features. The study is limited by the specificity of the OASIS dataset, which may affect the generalizability of the findings [7].

Bucholc et al. (2023) developed a novel machine learning (ML) framework using longitudinal data to predict the progression from MCI to dementia. A hybrid approach comprising unsupervised learning (to identify cognitive trajectory classes) and supervised learning (to develop prognostic models). National Alzheimer’s Coordinating Center Uniform Data Set (NACC-UDS). Included individuals with MCI diagnosis from the dataset. Two prognostic models were developed based on patient data from 3 visits (Model 1: n = 768) and 4 visits (Model 2: n = 409). Models incorporating cognitive trajectory classes showed up to 6.5% higher accuracy than models based only on clinical scores. The highest accuracy for Model 1 (data from three time points) was achieved by the ensemble and RF model, around 85%. For Model 2 (data from four time points), the highest accuracy was reported for RF and ensemble models, approximately 87.5%. Their study effectively utilized a small set of cognitive/functional assessments, inclusion of other data types like neuroimaging might enhance predictive capabilities [20]. Additionally, some existing studys’ methods and key limitations are mentioned below (see Table 1).

3. Methodology

This section outlines the methodology used in this research, including dataset description, data preprocessing, feature extraction, model development, evaluation metrics, and Explainable-AI. The overview of the methodology employed in this study is shown in Fig. 1.

3.1. Dataset description

The dataset utilized in this study offers an extensive perspective on the factors that contribute to the development of dementia. It covers a wide range of elements, including cardiovascular health, lifestyle selections, genetic indicators, demographics, and mental health. Notable variables within this dataset comprise alcohol consumption, heart rate, blood oxygen levels, and cognitive test scores. The dataset was gathered from Kaggle. Table 2 shows the features and their descriptions. This table provides a comprehensive view of the data features available in the dataset related to the study, with each row detailing a specific feature and its purpose or meaning. Each feature is labeled either as continuous, binary, or described without a specific data type when it is

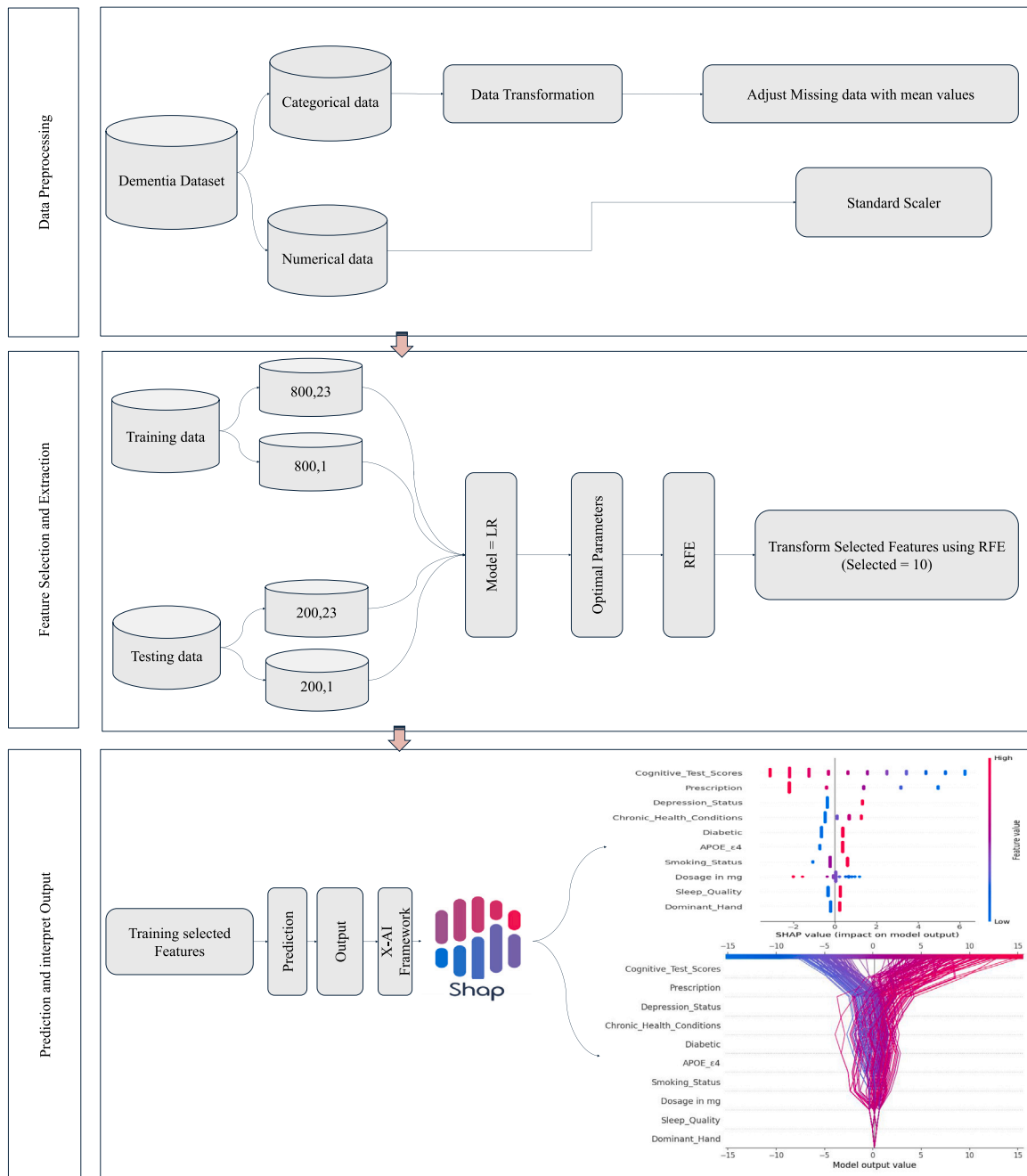


Fig. 1. Overview of the methodology employed in this study.

categorical or textual. This structure ensures clarity in understanding the type of data each feature represents and its relevance to the study. Moreover, The data underwent rigorous preprocessing, including standardization and feature selection, to ensure its suitability for machine learning applications. Although the dataset's attributes are comprehensive, some feature details may not have been fully elaborated in the original source, necessitating additional preprocessing steps in this study to enhance clarity and relevance.

However, The dataset contains data for a total of 1000 patients where 50.4% are female, and 49.6% are male. Fig. 3 shows the gender distribution among the dataset and Fig. 2 shows the age range of patients where patients age group between 60 to 90, (Dementia: 0 labeled as absence and 1 labeled as presence). However, the proportion of female patients suffering from dementia is slightly higher than that of male patients shown in Fig. 4.

3.2. Data preprocessing

3.2.1. Data cleaning

Data cleaning includes identifying and correcting errors within the data set. Thus, the Python library pandas data frame `data.dropna()` method was used to eliminate the rows that had `NULL` values. A feature was dropped because of the lower importance of the feature. However, all other features were highly correlated between each feature. Pandas data frame `data.isnull().sum()` method is used to check the missing values for each of the individual features in the dataset, which ensures that the dataset has zero null values.

3.2.2. Data transformation

Label encoding is a technique that is employed to simplify data by converting categorical variables into numerical values. In this particular research, a label encoding method was implemented to transform

Table 2
Dataset feature descriptions.

Feature	Description
Diabetes Status (Binary)	Indicates whether the patient has been diagnosed with diabetes (1 for yes, 0 for no).
Alcohol Consumption (Continuous)	Represents the patient's level of alcohol intake, which may provide insight into lifestyle habits.
Heart Rate (Continuous)	Number of heartbeats per minute, serving as a vital sign of cardiovascular health.
Blood Oxygen Saturation (Continuous)	Percentage of oxygen saturation in the blood, critical for assessing respiratory function.
Body Temperature (Continuous)	Patient's body temperature in Celsius, useful for identifying fever or hypothermia.
Body Weight (Continuous)	Patient's weight in kilograms, a basic health metric.
MRI Timing (Continuous)	The delay in receiving an MRI scan, which may indicate the severity or progression of conditions.
Prescribed Medication	Specific drugs prescribed to the patient, relevant for treatment tracking.
Medication Dosage (Continuous)	The prescribed dosage of medication in milligrams, important for dosage-effect analysis.
Patient Age (Continuous)	Age of the patient, a crucial demographic detail.
Educational Background	Highest educational level attained by the patient, possibly influencing health literacy.
Hand Dominance	Patient's dominant hand, potentially related to neurological factors.
Patient Gender	Gender of the patient, a basic demographic characteristic.
Family Dementia History	Whether there is a family history of dementia, a key risk factor.
Smoking Habits	Indicates the patient's smoking status, a significant lifestyle factor.
APOE ϵ 4 Allele	Presence of the APOE ϵ 4 allele, a genetic marker linked to Alzheimer's disease.
Exercise Frequency	The patient's level of physical activity, indicating lifestyle effects on health.
Depression Indicator	Reflects whether the patient has been diagnosed with depression, which may relate to cognitive health.
Cognitive Assessment Scores (Continuous)	Results from cognitive assessments, providing direct measures of cognitive ability.
Medication Use History	Tracks the patient's history of medication usage, important for evaluating drug interactions.
Dietary Habits	Describes the patient's dietary practices, a significant health factor.
Sleep Assessment	Evaluates the quality of the patient's sleep, a critical health aspect.
Chronic Conditions	Records any chronic illnesses the patient has, important for studying comorbidities.
Dementia Status (Binary)	The target variable, showing whether dementia is present (1) or absent (0).

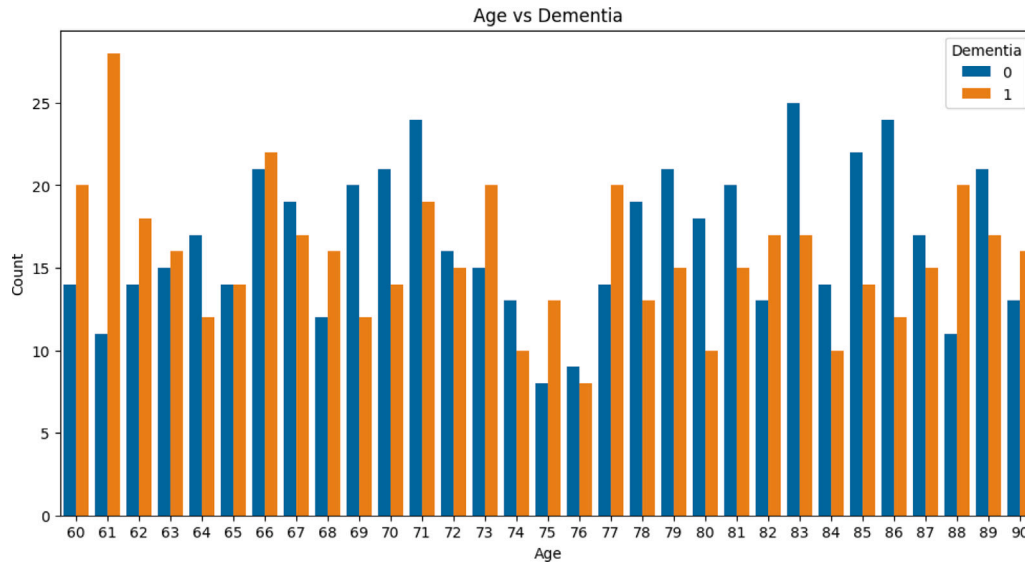


Fig. 2. Age group distribution in the dataset employed in this study.

the data from categorical variables to numerical values to analyze the data. However, the categorical feature variables were denoted as 0, 1, 2, 3, and so on, to facilitate the analysis process.

3.3. Feature selection

Feature selection is a technique that aims to reduce overfitting and underfitting and eliminate irrelevant features, thereby improving efficiency. In this study, several feature selection methods were utilized, including feature importance (Fig. 5) and correlation matrix with heatmap. Furthermore, the Seaborn library in Python was employed to visualize the correlation between each feature, allowing for the separation of highly correlated features and the removal of non-correlated features.

3.3.1. RFE

RFE is a feature selection technique that is well-suited for machine learning algorithms that assign weights to features, such as coefficients

in regression models. The objective of RFE is to choose features by recursively evaluating smaller sets of features. Initially, the estimator is trained on the initial set of features, and the importance of each feature is determined. Subsequently, the least important features are eliminated from the current set of features. This process is repeated recursively on the pruned set until the desired number of features is reached (Table 3). A figure is also illustrated to visualize the correlation among selected features in Fig. 6. This current study employed RFE to enhance the interpretable nature of our predictive model, minimize the likelihood of overfitting, and improve the model's capacity for generalization.

Chi-square test. The Chi-Square test statistic for this research is approximately 192.08, and its associated P -value is extremely small, about 1.12×10^{-43} . With only one degree of freedom, the results strongly reject the null hypothesis that the observed frequencies are equal to the expected frequencies [21,22].

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad (1)$$

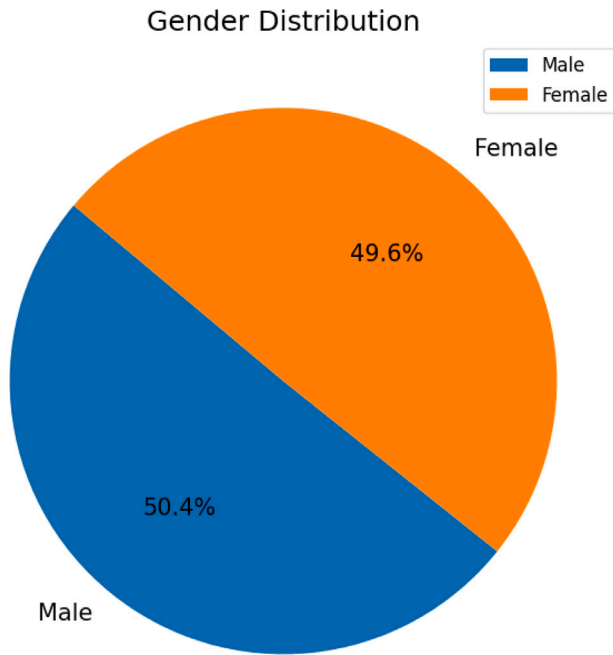


Fig. 3. Gender distribution in the dataset employed in this study.

Table 3
Extracted features using RFE.

Attributes	Data type	Values
Cognitive Test Scores	Numerical	0, 1, 2...
Prescription	Categorical	0, 1, 2, 3...
Chronic Health Conditions	Categorical	0, 1, 2, 3
Diabetic	Numerical	0, 1
APOEε4	Categorical	0, 1
Smoking Status	Categorical	0, 1, 2
Dosage in mg	Numerical	0, 1, 2, 3
Sleep Quality	Categorical	0, 1
Dominant Hand	Categorical	0, 1, 2, 3
Depression status	Categorical	0, 1

where O_i = observed value (actual value) and E_i = expected value.

3.4. Model train-test

In this study, the dataset was initially divided into training and testing sets, with 80% allocated for training and 20% for testing. This standard split ensures that the model can be trained on a sufficient amount of data while also being tested on a separate subset to evaluate its generalization performance. The target variable in this dataset represents the presence or absence of dementia in patients.

Moreover, to prevent model overfitting and enhance the generalizability of the results, Recursive Feature Elimination (RFE) was employed to select the most relevant features. By iteratively removing less significant features, RFE helped refine the feature set, ultimately reducing the dimensionality of the dataset. This preprocessing approach led to a more robust model by focusing only on the most influential features, ensuring that the model is not overly complex and that it can generalize well to new, unseen data.

3.5. Model selection

Selecting the appropriate model is a critical step in machine learning research that can significantly affect accuracy and efficiency. Choosing the ideal model can be overwhelming, but comprehending each model's advantages and limitations can assist in making an informed decision.

3.5.1. LR

LR is a statistical method used to predict the probability of a categorical outcome based on one or more predictor variables. LR is commonly employed to forecast binary outcomes, which are outcomes that can take on only two values, such as true/false, yes/no, or 1/0. LR is a valuable tool for solving classification problems [17,23–25].

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}} \quad (2)$$

$P(Y=1|X)$ is the probability of the event $Y=1$, given the predictor variable X . β_0 and β_1 are the parameters of the model. e is the base of the natural logarithm, approximately equal to 2.71828.

3.6. Proposed method

The LR model is a statistical method used for binary classification studies. In this study, a LR machine learning model is employed to analyze data in this study. However, a hyperparameter tuning regularization and novel RFE feature selection techniques are utilized to overcome the overfitting challenges. In Fig. 7, We have illustrated the method used in this study to overcome from over-fitting to perfect-fitting. The proposed method with RFE is also shown in Fig. 8 in detail.

3.7. Evaluation metrics

Evaluating the performance of a classification model, various metrics are required, such as accuracy Eq. (3) [17,26–28], precision Eq. (4) [29–31], recall Eq. (5) [32–34], and f1-score Eq. (6) [35,36]. Accuracy assesses the overall correctness of the model's predictions, while precision measures its accuracy in identifying positive instances. Recall evaluates the model's ability to capture all positive instances, and the F1-score provides a balanced score between precision and recall [10]. These metrics collectively offer a comprehensive understanding of the model's classification capabilities, enabling a nuanced assessment of its performance.

Mathematically, defined as:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$F1\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Where, TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative).

3.8. Confusion matrix

The confusion matrix (Fig. 9) provides a visual representation of the classification performance of the proposed model, showing the number of true positive, true negative, false positive, and false negative predictions. The proposed model achieved 94 true positives and 105 true negatives, accompanied by 1 false positive and 1 false negative. These figures facilitate the calculation of essential performance measures like accuracy, precision, recall, and f1-score, offering a thorough assessment of the model's proficiency in distinguishing between true and false classes. The results demonstrate a strong and efficient binary classification model.

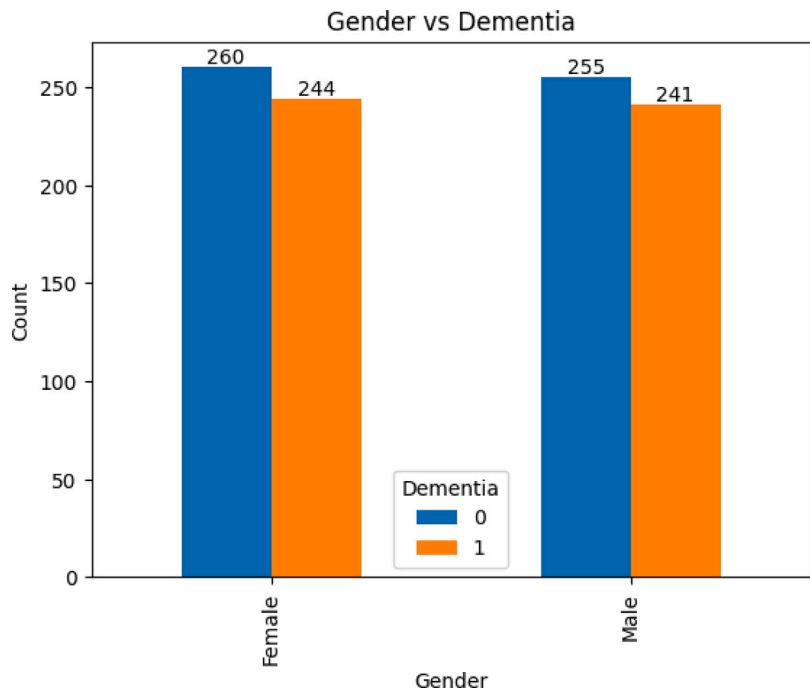


Fig. 4. Number of patients suffering from dementia.

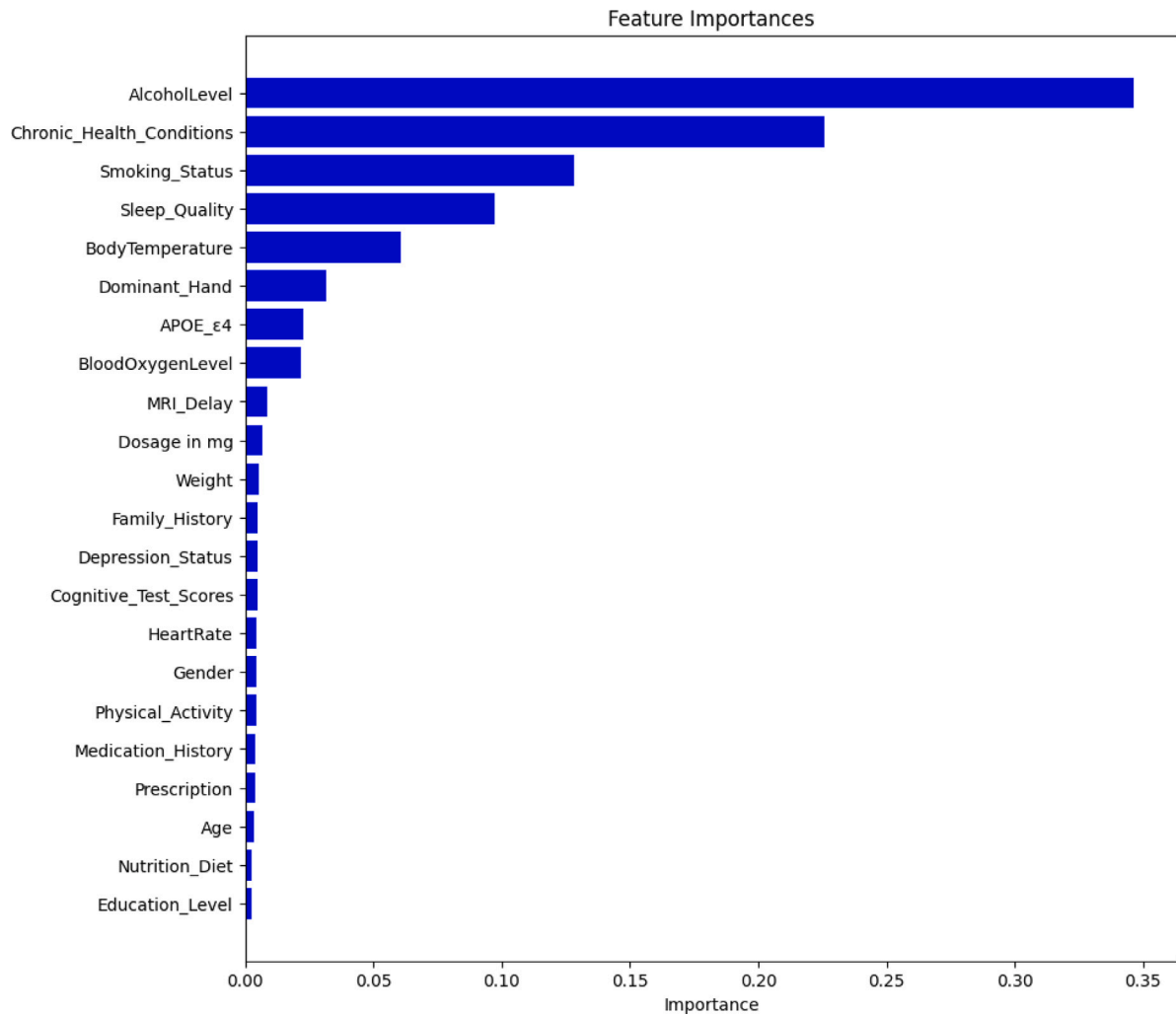


Fig. 5. Important features before applying RFE technique.

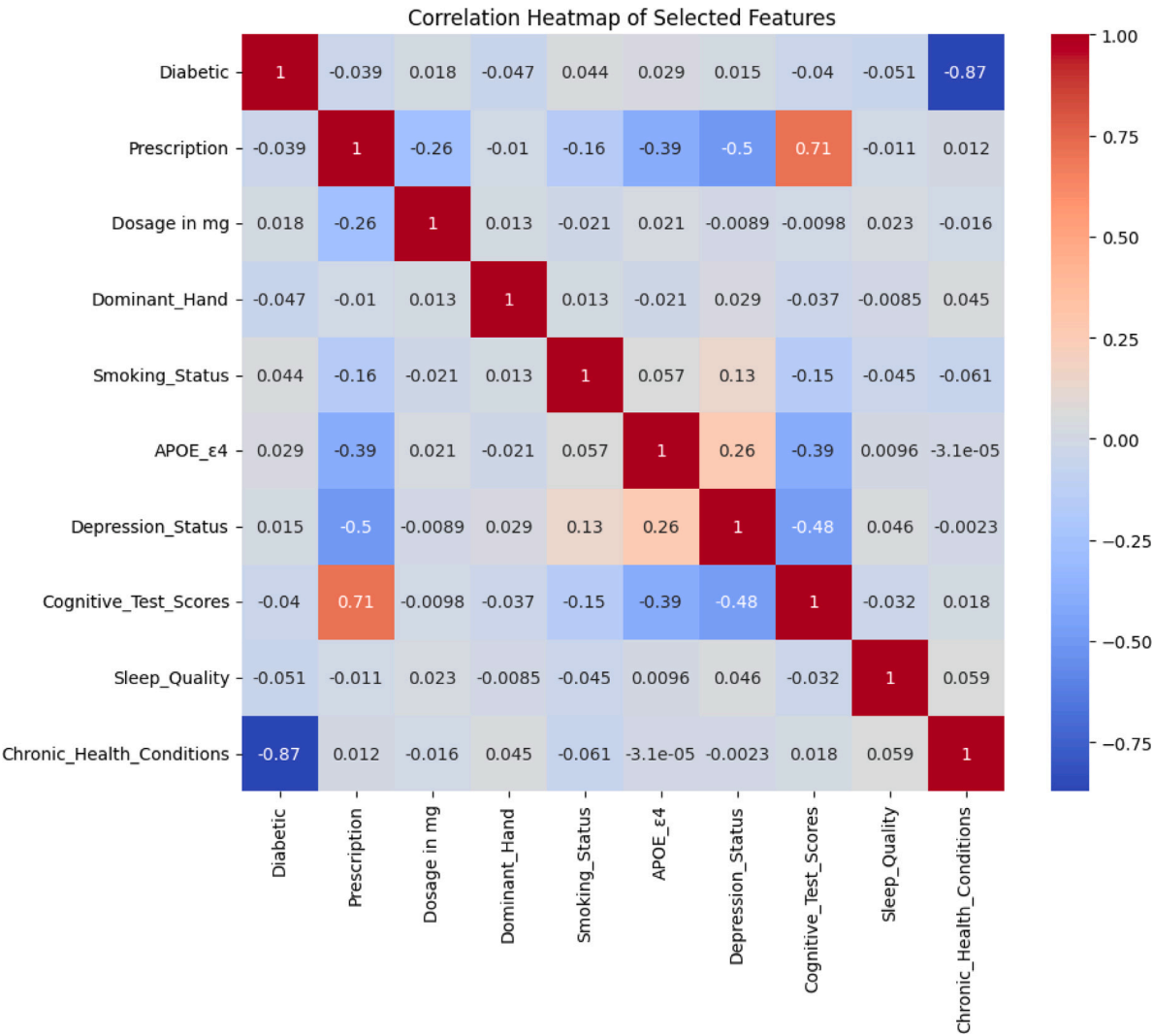


Fig. 6. Heatmap of features after applying novel selection techniques.

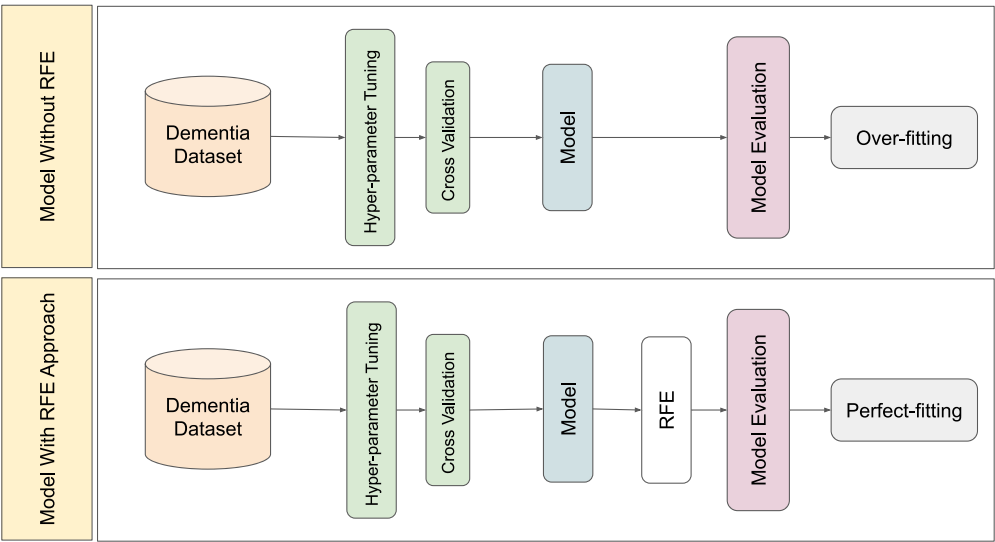


Fig. 7. Model with and without RFE method.

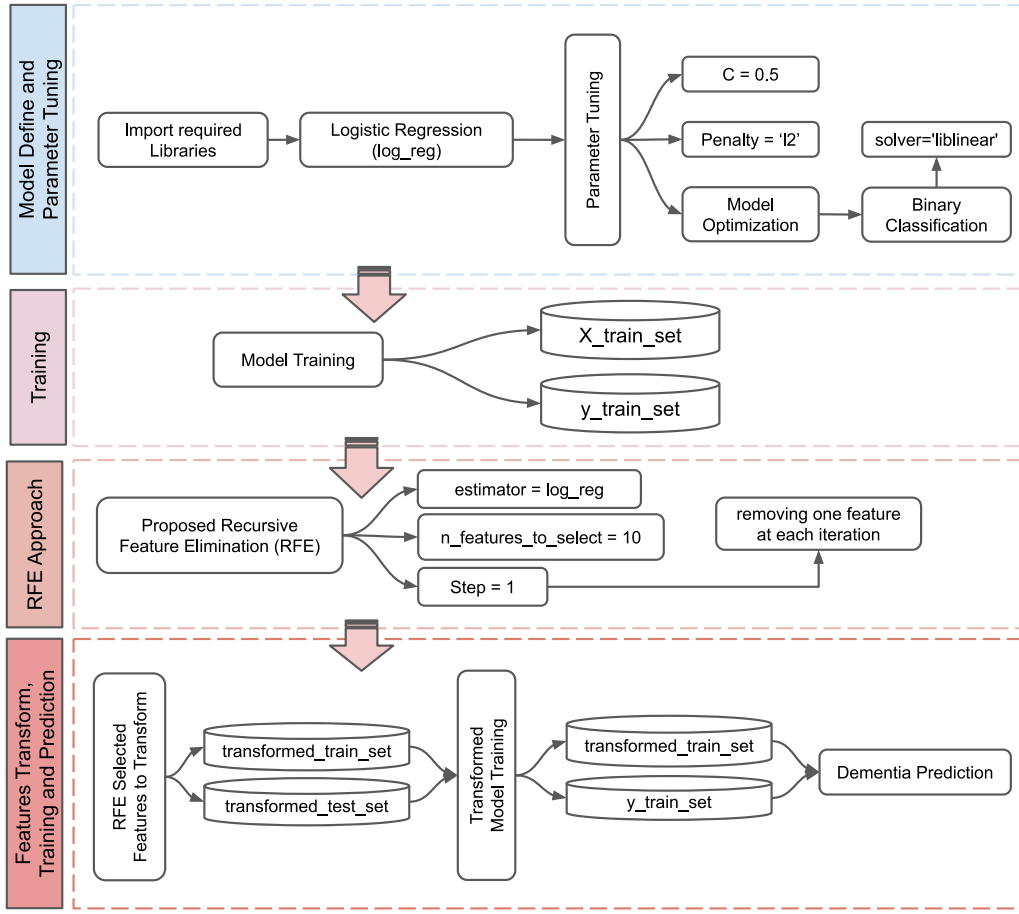


Fig. 8. Workflow of the proposed model with RFE.

3.9. Cohen's kappa coefficient

Cohen's kappa statistic is a widely used measure of agreement between two raters, particularly in assessing interrater reliability [37]. This statistical measure ranges from (−1) to (+1) in most correlation analyzes. The calculation of Cohen's kappa coefficient in this study was based on the confusion matrix (Fig. 9). The formula for Cohen's kappa coefficient is as follows [38]:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (7)$$

where P_o is the observed agreement and P_e is the expected agreement.

$$P_o = \frac{\text{Sum of Diagonal Elements}}{\text{Total Number of Instances}} \quad (8)$$

$$= \frac{94 + 105}{200}$$

$$\approx 0.995$$

$$P_e = \frac{\sum (\text{Row Total} \times \text{Column Total})}{(\text{Total Number of Instances})^2} \quad (9)$$

$$= \frac{(94 + 1) \times (94 + 0) + (0 + 105) \times (1 + 105)}{(200)^2}$$

$$= \frac{(95 \times 94) + (105 \times 106)}{(200)^2}$$

$$= 0.5015$$

Evaluating Cohen's Kappa Score:

$$\kappa = \frac{0.995 - 0.5015}{1 - 0.5015}$$

$$= \frac{0.4935}{0.4985}$$

$$\approx 0.9874$$

Cohen's Kappa score of 0.9974 indicates outstanding performance and a high degree of agreement between the model's predictions and the actual outcomes. Furthermore, the score suggests almost perfect agreement for this particular study.

3.10. MCC

The resultant MCC [39] of 0.9900 unequivocally demonstrates the model's remarkable predictive precision and dependability. This metric, encompassing a range from (−1) to (+1), evinces a strong positive correlation between the model's projections and the actual labels. The elevated MCC effectively corroborates the model's proficiency in effecting precise classifications. The results of this experiment provide substantial validation of the model's performance. Furthermore, MCC, defined as [40].

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}} \quad (10)$$

Where, TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative).

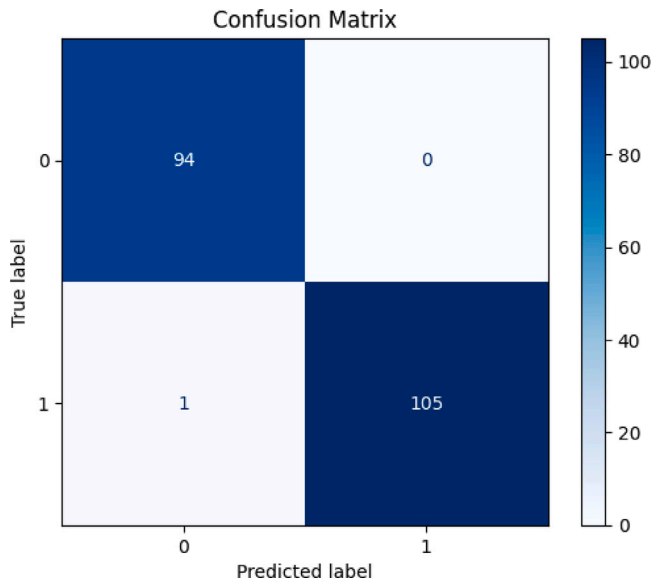


Fig. 9. Confusion matrix of the proposed model.

3.11. Experimental setup

The following tools were utilized for this study:

- Google Colab: As a platform provided by Google for the research community, Python was the only programming language used through a web browser [41].

In addition to the above application, the following Python libraries were employed:

- NumPy: A versatile package for manipulating multidimensional arrays and matrices in Python [42].
- Matplotlib: This cross-platform library was utilized in conjunction with NumPy to facilitate data visualization and graph plotting [43].
- Pandas: A software library was used to process and analyze data in tabular form [44].
- Scikit-learn: This comprehensive software library provides a wide range of tools for applying machine learning algorithms, including regression and classification algorithms [45].

4. Results

The results of our study indicate that the LR model with RFE significantly improves the prediction of dementia. This model achieved an accuracy of 0.995, a precision of 1.0, a recall of 0.991, and an F1-score of 0.995. Additionally, the MCC was 0.99, and the Cohen's Kappa score was 0.9874, indicating substantial agreement between the predicted and actual classes.

The LR model with RFE surpassed the standard LR model, confirming the efficacy of RFE in feature selection. According to the SHAP values (Figs. 17 and 18), cognitive test scores, prescription, depression status, chronic health conditions, diabetics, APOEε4, smoking status, dosages in mg, sleep quality and dominant hand were the most influential factors in predicting dementia. These findings align with previous research that identified these factors as significant predictors of dementia.

This proposed model of this current study is compared with other existing studies in Table 4 for predicting dementia. Our proposed LR with RFE method got higher accuracy 5 than others. Furthermore, this

Table 4

Previous Experiment result.

Citation	Method	Accuracy
Ortiz-Perez et al. (2023) [12]	CNN	90.36%
Franciotti et al. (2023) [13]	RF	90%
Kabir et al. (2023) [14]	KNN	96%

proposed model can predict dementia more efficiently (see Figs. 10 and 11).

The learning curve (Fig. 12) illustrates the model's performance as a function of the training set size on the x-axis and accuracy score on the y-axis. The cross-validation scores consistently ranged from 0.965 to 0.992, indicating strong performance. As the training set size (x-axis) increased, the training score (y-axis) rapidly improved, reflecting the model's ability to effectively learn and memorize patterns from the data. The plateau observed in the training score suggests that the model has reached its capacity to extract additional knowledge from the training data. Despite this, the stable and high cross-validation scores imply that the model generalizes well to unseen data, affirming its robustness in predicting outcomes on new datasets.

Fig. 13 illustrates where training accuracy remains consistently at 99.50%, while testing accuracy levels off at 99.38%, indicating that the model exhibits near-perfect learning on the training data but experiences a slight decline in performance when applied to unseen testing data. The constant 99.50% training accuracy suggests that the model has almost effectively memorized the training data. On the other hand, the 99.38% testing accuracy, although still exceptionally high, suggests that the model generalizes well but might encounter some challenges with a few instances in the testing set. Furthermore, Fig. 14 depicts that after using 10 epochs, the training loss steadily reaches 0.0335 while the testing loss remains below 0.0315 accuracy loss.

This study achieved an outstanding performance of the binary classification model, as evidenced by the ROC curve's value of 1.00 (Fig. 15). This outcome signifies perfect discrimination between classes, characterized by a true positive rate of 100% (sensitivity) and a false positive rate of 0% (specificity). This result demonstrates the model's exceptional ability to accurately distinguish between positive and negative instances without making any false predictions, which underscores its optimal discriminatory power and emphasizes its reliability in classifying instances.

The performance of the proposed classification model is accurately reflected in the precision-recall curve (Fig. 16), which demonstrates optimal results with 100% precision and recall. This suggests that the model can accurately identify positive instances with no false positives or negatives, a testament to its robustness and reliability in classification tasks.

In this research, Beeswarm plot shown in Fig. 17 illustrates the distribution of SHAP values across all selected input features, revealing their impact on model predictions. Notably, we observe that Cognitive_Test_Scores, Prescription, Depression, Chronic_Health_conditions emerge as top contributors. Decision plot shown in Fig. 18 further confirms the significance of these features, showcasing their cumulative impact on specific prediction outcomes. These SHAP plots offer insights into their importance which is shown in Fig. 19 in the model's decision-making process and enhance the interpretability of this study.

Additionally, in this study, SHAP were utilized as a posthoc explainable AI (XAI) method to provide insights into the contributions of individual features towards the model's predictions. The SHAP analysis is particularly valuable because it allows us to interpret the influence of each feature on the model's output, offering transparency into the decision-making process.

The application of SHAP in this context reveals the key factors that drive the prediction of dementia. For instance, cognitive test scores, prescription details, depression status, and chronic health conditions

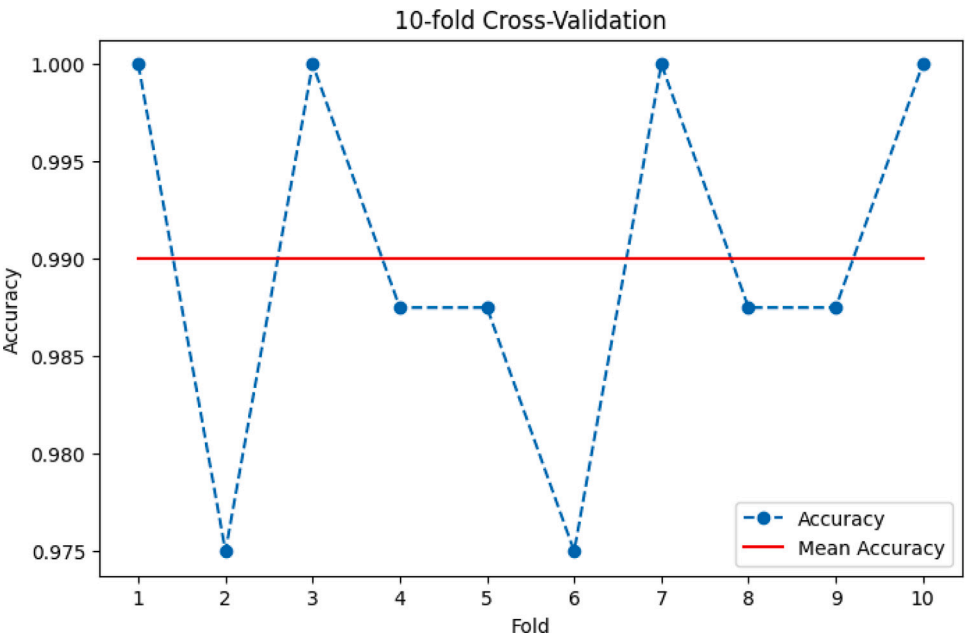


Fig. 10. 10-fold cross-validation employed in the proposed model.

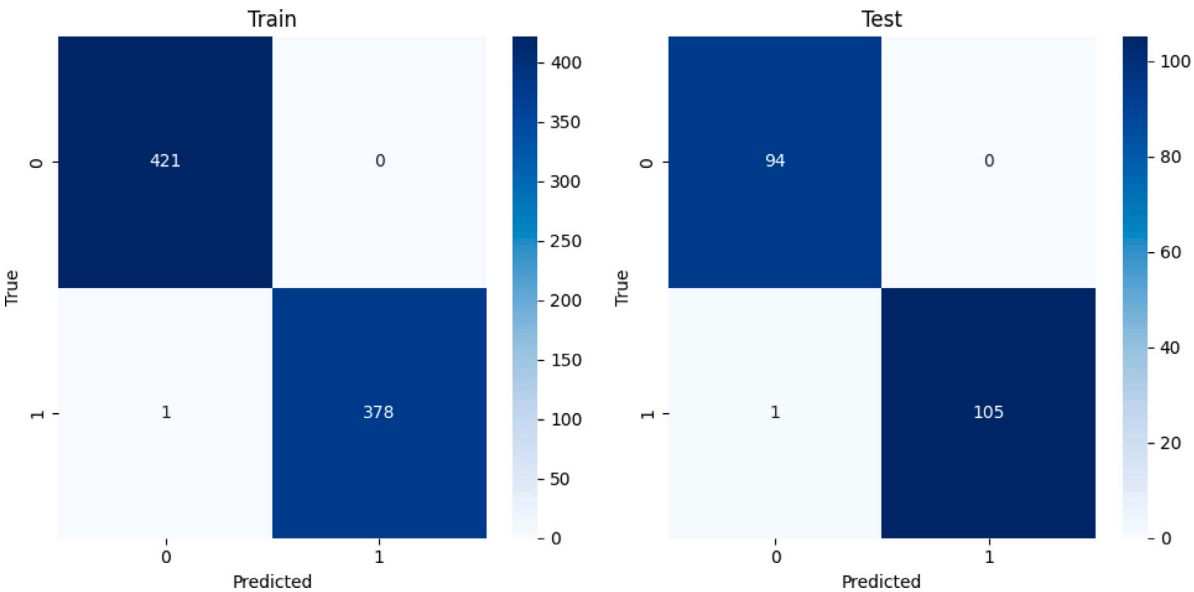


Fig. 11. Train test confusion matrix of the proposed model.

Table 5
Evaluation metrics of this experiment.

Model	Accuracy	Precision	Recall	F1-score	Cohens-Kappa	MCC	AUC
LR without RFE (Over-fitting)	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Proposed LR with RFE (Perfect-fitting)	0.995	1.000	0.991	0.995	0.9874	0.990	1.0

emerged as the most influential features. This aligns with existing medical literature, reinforcing the reliability of the model.

However, it is important to highlight that while SHAP provides a detailed breakdown of feature contributions, the manuscript lacked a thorough comparison with the feature importance derived from RFE, which was presented earlier. The RFE method, which iteratively removes the least important features and builds a model on the remaining

set, identified a similar set of important features, such as cognitive test scores and chronic health conditions.

Moreover, the motivation for including SHAP in our analysis was to complement the RFE results by not only identifying important features but also quantifying their impact on individual predictions. This dual approach allows us to cross-validate the importance of features identified by RFE, thus enhancing the robustness of our findings.

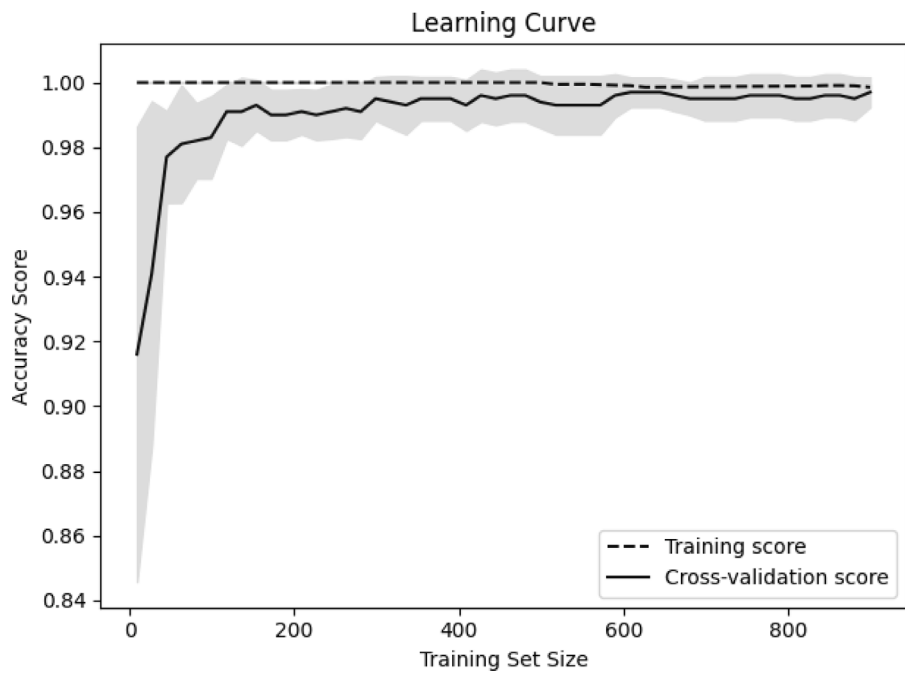


Fig. 12. Learning curve of this study.

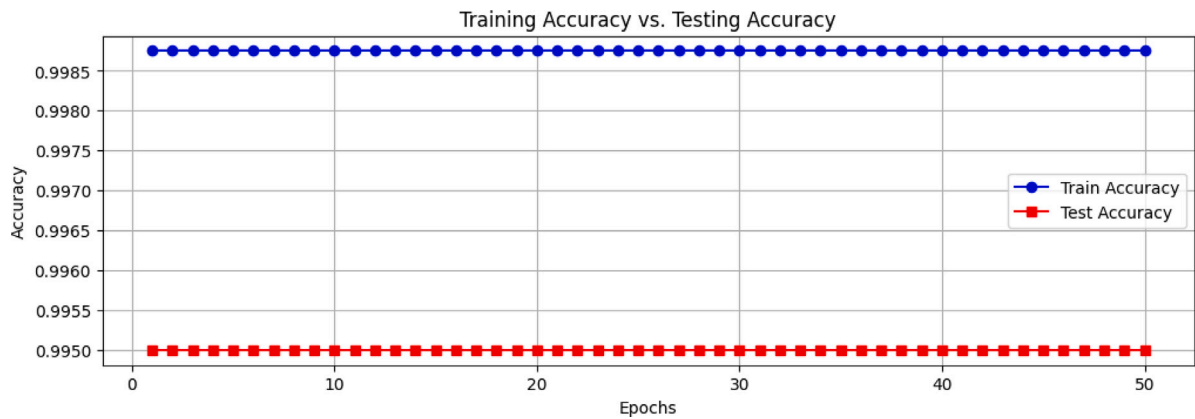


Fig. 13. Training and testing accuracy utilizing 50 epochs of this study.

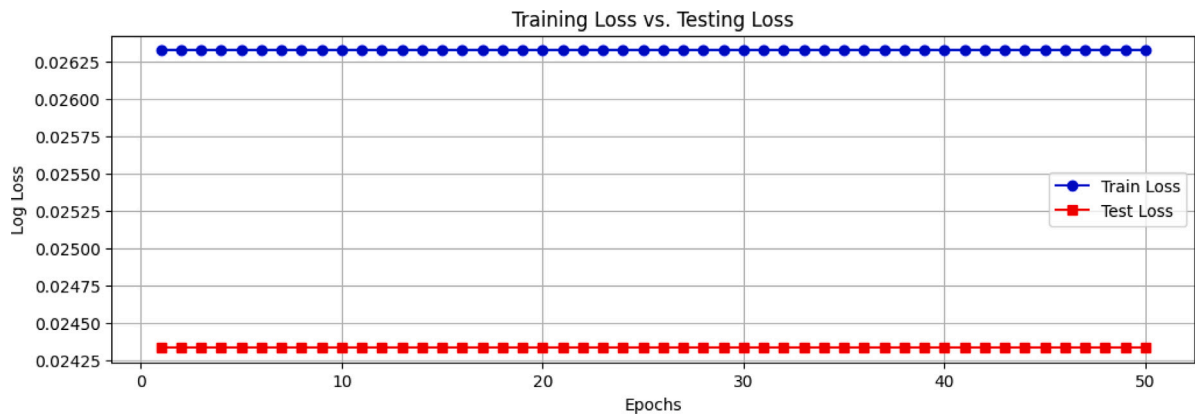


Fig. 14. Training and testing loss utilizing 50 epochs of this study.

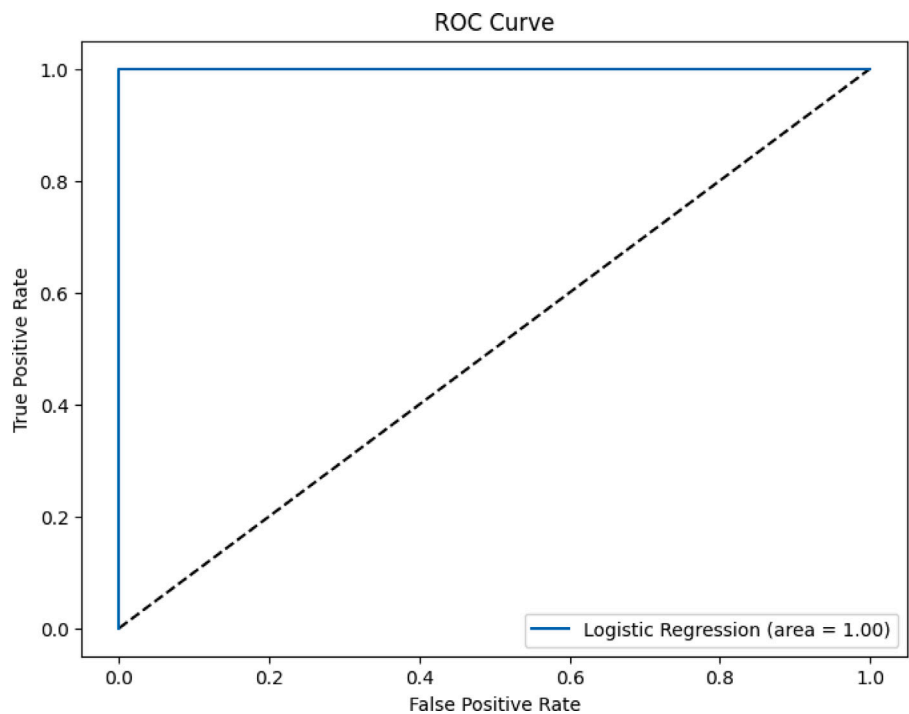


Fig. 15. ROC curve of this study.

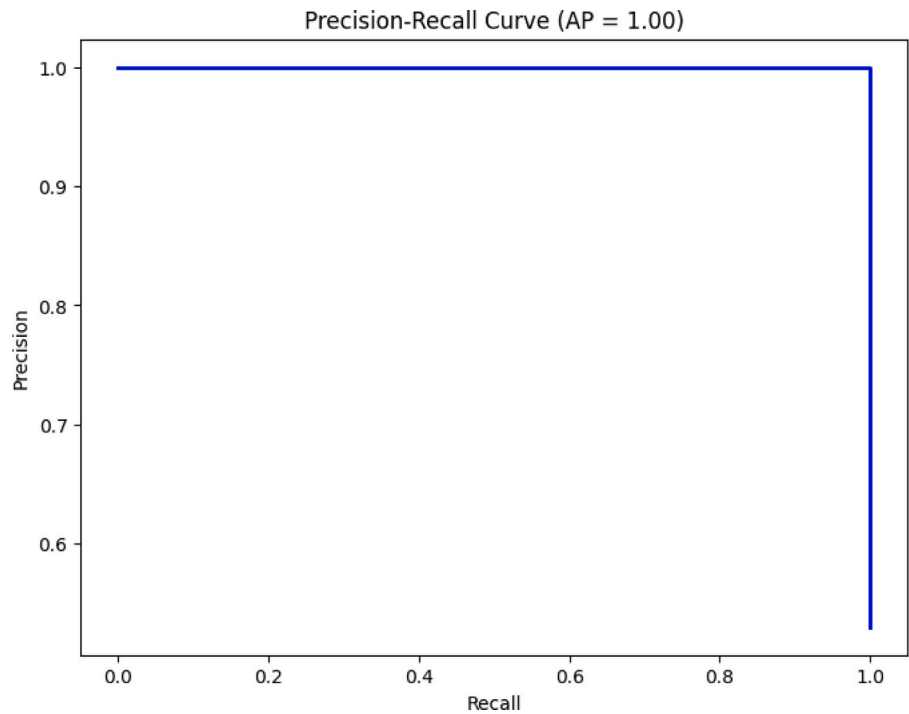


Fig. 16. Precision-recall curve of the proposed model.

In addition, upon comparing the results of SHAP with those obtained from RFE, we observed consistency in the key features identified, such as cognitive test scores and chronic health conditions. However, SHAP offers a more granular insight by showing the magnitude and direction (positive or negative) of each feature’s contribution to the prediction, which RFE does not provide. This comparison reinforces the validity of the selected features while offering a deeper understanding of how these features influence the model’s predictions.

Plus, the SHAP analysis provides readers with a clear understanding of the underlying factors that the model considers when predicting dementia. Furthermore, SHAP can help identify potential reasons for failed predictions, particularly by highlighting features that may contribute negatively to the accuracy of specific cases. However, it is essential to note that while SHAP adds interpretability, SHAP does not inherently improve the model’s performance, instead, SHAP offers an additional layer of transparency. The comparison with RFE underscores

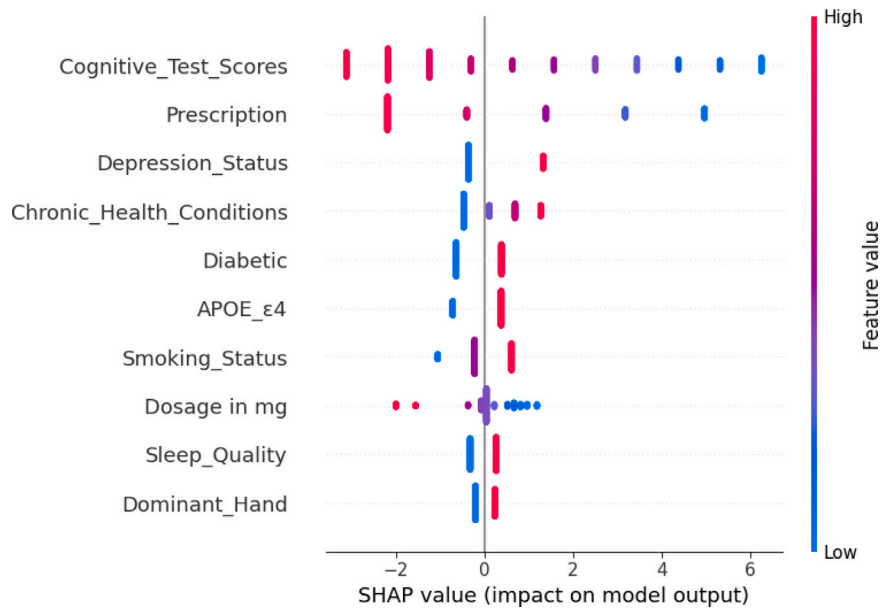


Fig. 17. Beeswarm plot of the proposed model.

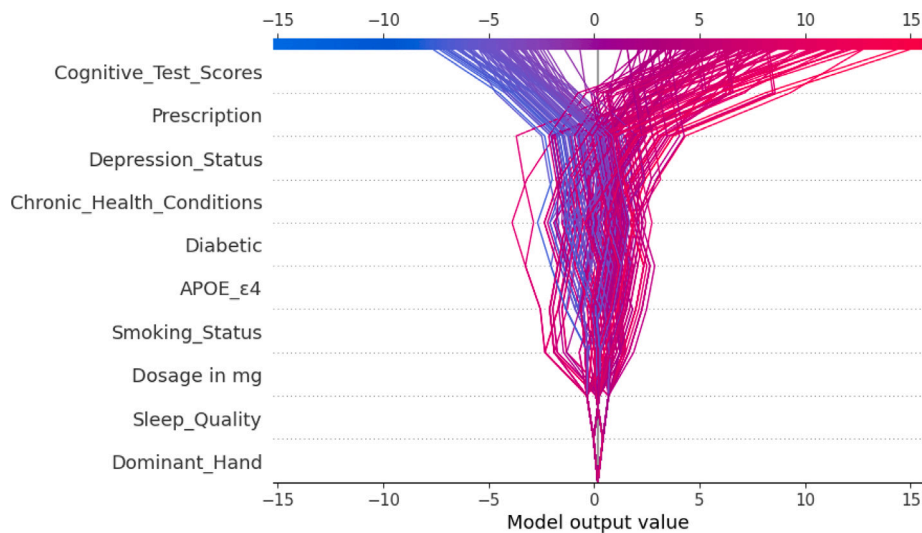


Fig. 18. Decision plot of the proposed model.

the importance of considering both feature selection and feature impact in model evaluation.

Additionally, The SHAP analysis conducted in this study provides crucial insights into the model's application for dementia prediction, as illustrated in Figs. 17, 18, and 19. These figures demonstrate the importance of various features in the model's decision-making process and offer a detailed breakdown of how these features influence individual predictions. The SHAP Beeswarm Plot (Fig. 17) highlights cognitive test scores as the most significant predictor, with higher scores correlating strongly with a higher likelihood of dementia, which aligns with the clinical understanding that cognitive decline is a primary indicator of dementia, thereby reinforcing the reliability of the model's predictions. Additionally, features such as prescription details, depression status, and chronic health conditions also significantly impact the model's output, suggesting that these factors should be closely monitored in clinical settings for early dementia detection. The Decision Plot (Fig. 18) supports these findings by visualizing the cumulative impact of these features on specific prediction outcomes, revealing how cognitive test scores, prescription details, and depression status interact

to shape the model's predictions. The SHAP feature importance bar plot (Fig. 19) consolidates this information, showing that cognitive test scores and prescription details have the highest average impact on the model's predictions, followed by depression status and chronic health conditions. Moreover, the SHAP analysis was instrumental in identifying outliers and misclassified cases, providing insights into potential model failures. For instance, one misclassified case exhibited high SHAP values for typically protective features, such as younger age or fewer chronic conditions, which misleadingly reduced the predicted dementia risk, highlighting the need for potential model adjustments, such as re-weighting features or incorporating additional data to address these edge cases. Overall, the SHAP analysis not only enhances the interpretability of the model but also offers practical insights into how healthcare professionals can leverage these findings to improve early dementia detection. By understanding the key factors driving the model's predictions, clinicians can better target their assessments and interventions, ultimately leading to more accurate and timely diagnoses.

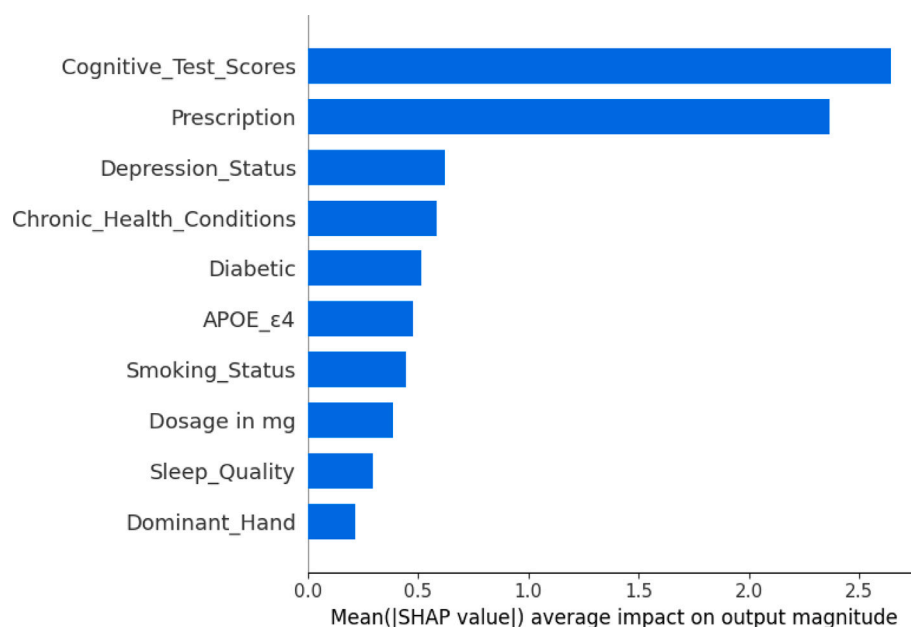


Fig. 19. SHAP feature importance bar plot of the proposed model.

5. Conclusion

The research presented a novel approach to predicting dementia using the LR model with RFE. The study demonstrated the effectiveness of the LR model with RFE in improving the accuracy and efficiency of dementia prediction. The use of SHAP values played a crucial role in enhancing the interpretability of the models, providing insights into the features that were most influential in predicting dementia. This not only improved the model's performance but also provided valuable insights for early prediction and intervention. Additionally, The findings of this research contribute to the growing body of knowledge in the field of dementia prediction and have potential implications for healthcare professionals and policymakers. By improving the accuracy and efficiency of dementia prediction, we can enhance patient care, optimize resource allocation, and ultimately, improve the quality of life for individuals affected by dementia. Moreover, Future research could explore other machine learning models and feature selection methods, as well as incorporate additional data sources for a more comprehensive analysis. Furthermore, the interpretability of the models could be enhanced to provide more detailed insights into the prediction process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset for this study is available at the following link. [Dataset \[9\]](#).

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